STATISTICAL ANALYSIS & MACHINE LEARNING REPORT

UCI DATASET ID: 144

German Credit Risk Prediction

GROUP 3

TARGET: Statlog German Credit Data

RESULT: 44.3% Reduction in Total Risk Cost



OPTIMIZED DECISION BOUNDARY

RESEARCH TEAM

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DEMO

EXECUTIVE ROADMAP

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CONTEXT & PROBLEM STATEMENT

⚖️ The Credit Asymmetry & Accuracy Paradox

CANONICAL LOSS PENALTY RATIO

5 : 1

Cost(False Negative) = 5 • Cost(False Positive) = 1

False Negative Damage (FN = 5)

Granting credit to a defaulting applicant incurs direct financial loss equal to loan principal default (Exposure at Default × Loss Given Default) plus legal recovery costs.

False Positive Penalty (FP = 1)

Rejecting a creditworthy applicant incurs an opportunity loss consisting of foregone net interest margin and customer relationship lifetime value.

The Accuracy Paradox: Standard machine learning classifiers tuned to maximize accuracy at a default 0.50 cutoff achieve ~75% accuracy but suffer disastrous financial losses (0.853–1.006 cost/customer), performing worse than a naive "Reject All" policy (0.700 cost/customer).

DATA INTEGRITY & HYGIENE

UCI Statlog Provenance & Grömping (2019) Audit

DATASET BENCHMARK

1,000 Records

700 Good (70%) vs 300 Bad (30%)

DATA COMPLETENESS

100% Complete

0 nulls, 0 duplicate profiles

FEATURE TAXONOMY

20 Attributes

7 numeric, 13 categorical

Historical Coding Corrections (Grömping, 2019 Research Audit)

Attribute	UCI german.doc Definition	Original 1973 German Bank Coding Correction
Attribute 3 (credit_history)	A30='no credits', A34='critical'	Scrambled! A30=past delays (62.5% Bad), A34=critical/paid duly (17.1% Bad).
Attribute 20 (foreign_worker)	A201='yes' (963 rows)	Inverted binary labels relative to German original documentation text.
Attribute 18 (num_dependents)	Code 1 vs 2 tiers	Swapped: Tier 1 = ≥3 dependents (15.5%), Tier 2 = 0-2 dependents (84.5%).

Source: UCI Machine Learning Repository - Statlog (German Credit Data) / Grömping, U. (2019), "South German Credit Data".

EXPLORATORY DATA ANALYSIS

Numerical Attributes & Non-Linear Hazard

duration_months (Attr2)

p < 0.0001

Median: Good 18m vs Bad 24m (+33.3%). Loans >36m default at a severe **52.0% rate**.

credit_amount (Attr5)

Skew: +1.95

72 high-value outliers (>7,882 DM) exhibit a **54.0% Bad rate** (vs 30% baseline).

age (Attr13)

p = 0.0004

Young borrower hazard: Applicants ≤25y default at **42.0% rate** vs 24.0% for 36-55y.

Mann-Whitney U Non-Parametric Significance Summary

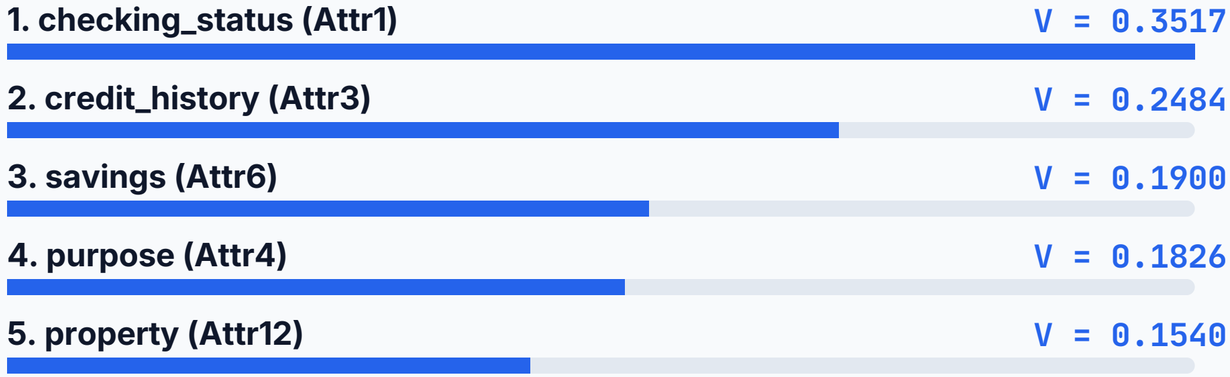
Numerical Attribute	Good Median (n=700)	Bad Median (n=300)	p-Value	Statistical Status
duration_months	18.0 months	24.0 months	< 0.0001	**Highly Significant**
age	34.0 years	31.0 years	0.0004	**Highly Significant**
credit_amount	2,244.0 DM	2,574.5 DM	0.0059	**Significant**
existing_credits / num_dependents	1.0 tier	1.0 tier	> 0.1300	Not Significant

Source: UCI Statlog German Credit Dataset Mann-Whitney U Hypothesis Tests (α = 0.05).

BIVARIATE RISK SIGNALS

Categorical Feature Association Ranking

Cramér's V Association Ranking



Source: Chi-Square Independence Tests & Cramér's V Coefficients.

checking_status Dominance

Applicants with **< 0 DM** balance default at **49.3%** (n=274) vs **11.7%** for applicants with no checking account (n=394).

Savings & Purpose Divergence

Savings < 100 DM defaults at **36.0%** vs **12.5%** for ≥1,000 DM. Purpose education defaults at **44.0%** vs **16.5%** for used cars.

Weak Bivariate Predictors

Job category (p=0.5966) and telephone ownership (p=0.2789) exhibit statistically insignificant default variation.

METHODOLOGY & PIPELINE

Leakage-Free Experimental Architecture

Step 0

Partition Isolation

800 Train / 200 Test split (SEED = 42). Preserves exact 30.0% default rate.

Step 1

Leakage Audit

Single-tree AUC max 0.700 (<0.85). Mutual Info max 0.0608 (<0.10). 0 duplicates.

Steps 2-5

Decoupled Tuning

GridSearchCV on tune_cv (SEED = 43) refitted on full 800 training rows.

Steps 6-8

OOF Calibration

OOF probabilities generated on skf (SEED = 42) to optimize threshold t*.

Zero Test Contamination Guarantee

Scalers, OneHotEncoders, hyperparameter selection, and threshold calibration are strictly restricted to training folds. The 200-row test set is accessed exactly once in Step 8 to evaluate final holdout performance.

FEATURE ENGINEERING

ColumnTransformer Pipeline Architecture

1 Log-Normalizing Branch

`['credit_amount']`

Applies FunctionTransformer(np.log1p) followed by StandardScaler().

Mitigates +1.95 Skew

2 Standard Numeric Branch

`[7 Numeric Attributes]`

Applies StandardScaler() directly to duration, age, installment rate, etc.

Zero Mean, Unit Variance

3 Regularized OHE Branch

`[13 Categorical Columns]`

OneHotEncoder(min_frequency=30, drop='if_binary'). Groups sparse levels.

Yields 56 Feature Dimensions

DECISION THEORY MATH

Asymmetric Cost Function & Bayes Optimal Threshold

TOTAL PORTFOLIO MISCLASSIFICATION LOSS FUNCTION

$$\text{Total Cost}(\tau) = 1 \times FP(\tau) + 5 \times FN(\tau)$$

$$\text{Avg Cost per Applicant}(\tau) = [1 \times FP(\tau) + 5 \times FN(\tau)] / N$$

Bayes Theoretical Cutoff

$$p^* = C_{FP} / (C_{FP} + C_{FN}) = 1 / 6 \approx 0.167$$

Predict 'Bad' (Reject) if default probability ≥ 0.167.

Trivial Benchmark Floor

Reject All Policy = 0.700 Cost / Cust

Approve All = 1.500 cost/cust. Any model must beat 0.700.

BASELINE EVALUATION

Untuned Model Performance at Default Cutoff 0.50

Model Architecture	Pooled OOF AUC	Cost 0.50 Cutoff	Cost 1/6 Bayes	Optimal OOF t*	Cost/Customer Optimal
XGBoost (Untuned)	0.789	**682 (0.853/c)**	448	0.10	**0.509**
Random Forest (Untuned)	0.793	**805 (1.006/c)**	430	0.14	**0.530**
Logistic Regression	0.768	**730 (0.913/c)**	438	0.12	**0.541**
SVM (RBF Kernel)	0.779	**752 (0.940/c)**	457	0.19	**0.559**
KNN / Neural Network	0.700 - 0.734	**> 1,035 (>1.29/c)**	476 - 522	0.17 - 0.19	**0.595 - 0.599**

Source: 5-Fold Stratified Cross-Validation Out-of-Fold (OOF) Baseline Evaluations (N=800).

OPTIMIZATION & CALIBRATION

Hyperparameter Tuning & OOF Threshold Convergence

Post-Tuning OOF Discrimination Gains

XGBoost (n_est=150, lr=0.05, depth=3)	AUC 0.796 (+0.007)
Logistic Regression (C=0.03 L2)	AUC 0.778 (+0.010)
Neural Network (hidden=32, lr=0.01)	AUC 0.752 (+0.052)
KNN (k=21, Manhattan p=1)	AUC 0.771 (+0.037)

Empirical Threshold Band Convergence

Sweeping candidate thresholds (0.01 to 0.99) on OOF predictions shifts optimal decision boundaries into **0.14 - 0.21**.

Theoretical Alignment

Empirical range 0.14–0.21 closely tracks the Bayes theoretical cutoff 1/6 (≈0.167), proving that retaining natural 30% training balance preserves calibration.

OUT-OF-SAMPLE RESULTS

Champion Model Performance on Holdout Test Set (N=200)

Champion Architecture

XGBoost (Threshold = 0.15)

0.535
Test Cost / Cust (23.6% Savings)

Model	Locked t*	Conf Matrix (TN/FP/FN/TP)	Test Cost/Cust	Recall	Test AUC / Gini
1. XGBoost	0.15	63 / 77 / 6 / 54	**0.535**	**90.0%**	0.803 / 0.606
2. Random Forest	0.14-0.21	41 / 99 / 2 / 58	**0.545**	**96.7%**	0.796 - 0.800
3. Neural Network	0.14-0.21	86 / 54 / 11 / 49	**0.545**	81.7%	< 0.796
4-6. LR / SVM / KNN	0.14-0.21	42 / 98 / 6 / 54	**0.550 - 0.640**	90.0%	Lower Gini

Source: Independent Holdout Test Set Evaluation (N=200, Default Ratio = 30.0%).

UNCERTAINTY ANALYSIS

2,000-Iteration Bootstrap 95% Confidence Intervals

95% Bootstrap Cost Bounds

XGBoost (Champion)
Entire 95% bound remains strictly below 0.700 cap.

`[0.415, 0.665]`

Random Forest
Extremely stable risk capture across resamples.

`Upper CI ≤ 0.640`

Neural Network / LR
Wider confidence band due to calibration sensitivity.

`Upper CI ≤ 0.740`

Sample Size Granularity Effect

On a 200-sample test set (containing 60 defaults), a **single missed defaulter (FN)** shifts average cost per applicant by **0.025 units**.

This granular shift causes bootstrap CIs to overlap among top contenders, making tree ensemble stability crucial for production.

FEATURE SELECTION ANALYSIS

Permutation Importance & Nested Top-k Ablation

Top Predictors (Validation Drop)

1. checking_status (Attr1) — ΔAUC = 0.0970
2. duration_months (Attr2) — ΔAUC = 0.0520
3. savings (Attr6) — ΔAUC = 0.0242
4. credit_history (Attr3) — ΔAUC = 0.0216
5. credit_amount (Attr5) — ΔAUC = 0.0214

Nested Top-k Feature Ablation Study

Features (k)	OOF AUC	AUC Change	Cost/Cust @Opt
k = 20 (Full)	**0.796**	**0.000**	**0.506**
k = 15	0.784	-0.012	0.539
k = 10	0.780	-0.016	0.555
k = 5	0.756	-0.040	0.576
k = 3	0.725	-0.071	0.562

× Pruning features monotonically degrades discrimination and increases loss. Retain full 20 features.

BANKING DEPLOYMENT ARCHITECTURE

3-Zone Traffic-Light Policy & Compliance Rules

Green Zone

`p_hat < 0.14`

Straight-Through Processing (STP) Auto-Approval for high-quality, low-hazard credit applicants.

Amber Zone

`0.14 ≤ p_hat < 0.21`

Manual Underwriting Review / Require Guarantor or Term Cap adjustments before final approval.

Red Zone

`p_hat ≥ 0.21`

Automated Decline to prevent high-cost default exposure and protect loan portfolio quality.

EU AI Act & GDPR Article 22 Compliance Directives

Classified as High-Risk AI under Annex III §5b. Integrate TreeSHAP for adverse-action notice generation. Remove protected demographic attributes (personal_status_sex Attr9, foreign_worker Attr20) prior to production deployment while maintaining ROC-AUC within 0.001–0.012 of benchmark.

RESEARCH CONCLUSIONS & SUMMARY

🏁 Methodological Delivery & Key Findings

1. Resolution of Accuracy Paradox

Proved default 0.50 cutoff models fail financially. Threshold optimization to 0.14–0.21 reduces misclassification loss by 23.6% (saving 0.165 cost units per applicant).

2. Champion Architecture Established

XGBoost achieved Rank 1 test cost (0.535), 0.803 ROC-AUC, 0.606 Gini, and 90.0% Recall, maintaining entire 95% bootstrap CI below the 0.700 benchmark.

3. Feature Retention & Data Hygiene

Nested top-k ablation verified that all 20 attributes are required. Resolved historical 1973 German credit coding errors (Grömping, 2019).

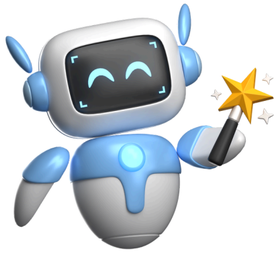
4. Fully Reproducible Delivery

Delivered 100% leakage-free Python 3.13 codebase (`Project_Statistics_full_final.ipynb`), complete data dictionary, and EU AI Act compliant routing.



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Product Demo



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